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Librarians' views of the readiness of university libraries in Indonesia to adopt virtual and augmented reality

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Abstract

Purpose – This study aims to determine the extent of virtual reality (VR)/augmented reality (AR) adoption in university libraries in Indonesia and identify librarians' views of library readiness. The assessment centers on two crucial dimensions: library readiness, gauged by commitment and efficacy, and librarian readiness, appraised through cognitive and affective aspects.

Design/methodology/approach – The research features a survey-based quantitative method. A sample of 259 librarians from university libraries, all members of the Indonesian Higher Education Library Forum, responded to the online survey.

Findings – Indonesian university libraries, especially in Central Java, express strong interest in VR/AR despite low current adoption. Librarians believe in their commitment and efficacy for adoption, with improvements needed in budget procurement and human resources. Librarians are optimistic about innovating academic activities with VR/AR, despite concerns about collaboration, data governance and human resource insecurity. The research underscores the simultaneous influence of librarians' and libraries' readiness on VR/AR adoption, emphasizing their interconnectedness and mutual support within library management.

Practical implications – There are practical implications for library institutions in Indonesia in terms of understanding the attendant opportunities and challenges when preparing organizational ecosystems. These findings provide a broad overview of VR/AR adoption in university libraries in Indonesia and contribute to understanding the specific needs and interests of universities in Indonesia considering adopting VR/AR. In addition to preparing adequate infrastructure, libraries should ensure the availability of adequate human resources.

Originality/value – In the Indonesian context, this study, to the best of the authors' knowledge, makes the first attempt to examine the librarian's view as part of library management to face the dilemmas that may arise during the adoption of VR/AR despite university libraries' commitment and

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efficacy to enable this technology. This research also identifies librarians' cognitive and affective readiness as individual characteristics and resources able to influence adequate adoption.

Keywords Library readiness, Librarian readiness, Virtual reality, Augmented reality, University libraries

Paper type Research paper

Introduction

Libraries must navigate challenges arising from new technologies, notably virtual reality (VR) and augmented reality (AR). According to Ellern and Cruz (2021), libraries must devise strategies to actively participate in shaping individuals' collective future, whether it be "real, virtual, or both". Noteworthy VR/AR integration in academic library services includes the use of library tours and the Association of Research Libraries' incorporation of Oculus Rift (Sample, 2020), High Tech Computer (HTC) Vive for digital scholarship (Greene and Groenendyk, 2020), myLibrARy to locate books and information within the library, Special Collection using Augmented Reality to Enhance Learning and Teaching (SCARLET) and Virtual Learning Center (VLC) (Roy *et al.*, 2022).

Libraries have begun experimenting with immersive technologies, although the extent to which they are doing so is unclear (Greene and Groenendyk, 2020). While some libraries have embraced VR/AR, a substantial number still grapple with barriers to adoption (Pope, 2018). In Indonesia, the implementation of VR/AR in libraries continues to lag. In July 2019, Soegijapranata Unika Semarang library launched VR/AR as "the first library that uses the Augmented Reality (AR) device in Indonesia" (Unika, 2019). Shortly after, Unila library, Lampung, launched an innovative library service based on VR/AR that allows visitors to see facilities virtually and view the building using a virtual tour (VT) application and VR devices, alongside *Augmented Reality for Library* technology, which has a navigation function to access various library features (Unila, 2019).

Integrating VR/AR into library services represents a paradigm shift, demanding that librarians be at the forefront of technological advancements to enrich user experiences (Newman and Newman, 2021). Successful integration requires librarians to enhance their knowledge, skills and digital literacy (Subaveerapandiyan *et al.*, 2022). Evaluating the cognitive and affective readiness of librarians, alongside the commitment and efficacy of their institutions (Adeyemi and Omopupa, 2020), is imperative to gauge the readiness of libraries to adopt VR/AR.

This study identifies challenges such as infrastructure availability, ICT skills and technophobia, drawing on Shonhe's (2019) use of the Technology Readiness Index (TRI) to assess librarians' readiness for technology adoption. Similarly, it evaluates the readiness of Indonesian university libraries to function as VR/AR service providers. Assessing librarians' cognitive and affective aspects is crucial to understanding their ability to manage libraries in the digital era.

Despite the potential benefits, to date, there has been little research on VR/AR in Indonesian university libraries. This study aims to fill that gap by investigating the extent of VR/AR technology adoption, utilization patterns and the reasons for slow progress. The findings will guide university libraries and stakeholders in crafting change management strategies, devising training programs and allocating resources to ensure the effective incorporation of VR/AR, particularly in advancing digital scholarship.

In 2019, Shonhe used the TRI to assess librarians' readiness to adopt library management and information dissemination technology. The author concluded that the main challenges were the need for increased availability and maintenance of infrastructure, the lack of information, communication and technology (ICT) skills and technophobia or fear on the part of

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librarians of using new technology. Similarly, this study sought to determine libraries' readiness to adopt VR/AR.

How many university libraries in Indonesia have adopted VR/AR and how they have used it needs to be determined, as well as the reasons for slow adoption. This study was developed to answer these questions. It scrutinizes management dimensions such as commitment and efficacy while delving into the cognitive and affective aspects of librarians' readiness. By addressing these facets, the study aims to contribute valuable insights for effectively integrating VR/AR into academic functions, enhancing digital scholarship and positively impacting the Indonesian educational landscape.

Literature review

Virtual reality/augmented reality in university libraries

VR/AR represents a group of technologies that aim to supplement "the real world" via increasingly immersive computer-generated content (Grant and Rhind-Tutt, 2019). VR/AR spans a gradual spectrum, anchored by the real world at one end and a wholly computer-generated world at the other extreme. According to Lischer-Katz and Cook (2022), the main characteristics of VR are immersion, interactivity and visualization.

VR/AR applications enable users to manipulate virtual environments in several ways, such as via a handheld controller or eye tracker controls. Visualization allows objects to be displayed in a VR environment as 3D objects with properties like the original. There are various applications of VR/AR in university libraries designed to help libraries and librarians support academic activities, including the following (Aulianto, 2020):

(1) Library tour

A practical scheme for navigating the library and locating specific collections. Ng *et al.* (2023) and Chu demonstrated that CoSpaces, a VR platform, enhances language learning through a 3D model library that supports VR tools to explore digital books and library tours, which foster digital literacy in various dimensions of knowledge across context and skills.

(2) Library services

VR/AR in libraries provides four functions: (a) present three-dimensional location directions; (b) offer visual information that leads to the required collection; (c) enable voice search capability; and (d) allow users to explore virtual digital content in a real environment (Su and Chen, 2022).

(3) Information literacy

For Hicks and Lloyd (2021), information literacy is divided into skills-based, a skill that is learned so that it can be applied in other contexts, and practice-based, which emphasizes the importance of the context in which valuable ways of carrying out an action are determined. VR/AR literacy materials are accessed with VR equipment, either via content in proprietary information literacy modules or by exploring material already widely available online (Aulianto, 2020).

(4) Online catalog

Ternenge *et al.* (2021) investigated the perceived impact of using OPAC, the online catalog in the library, to search for information effectively. VR/AR in an online catalog system provides unique experiences tailored to the user, for example, guiding users virtually to the desired collection rack before they physically move (Aulianto, 2020).

(5) Digital scholarship

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VR provides a powerful digital scholarship tool. This ever-changing creative field uses digital tools and techniques, such as data visualization, text analysis, mapping and scholarly multimodal publishing, to extend traditional research methods (Aulianto, 2020).

Library readiness

Library readiness describes the ability to respond to management changes by creating profits, minimizing risks and maintaining performance. Rusly *et al.* (2015) described this concept as an intervention to minimize resistance to the technological developments that have changed the nature of library services, workflow and readiness. The constantly shifting technological landscape challenges and influences libraries to optimize resources to meet users' needs. For Sayogo *et al.* (2020), digital transformation in libraries challenges aspects such as library qualifications, capabilities, funds and resources, rules and responsibilities and leadership commitment.

Evidence suggests a readiness mindset can increase an organization's adaptability to change (Bell, 2019). Weiner (2009) states that organizational readiness is characterized by a collective psychological state within the organization, marked by the commitment of individuals to implement change (change commitment) and their confidence in the collective abilities of the organization to successfully bring about that change (change efficacy). This underscores the importance of fostering a positive and committed mindset among organizational members to facilitate successful organizational change initiatives.

The *Organizational Readiness for Adopting Change (ORIC)* scale, which measures an organization's readiness to change, was developed based on Weiner's theory of organizational readiness for change (Shea *et al.*, 2014). The scale's questions interrogate two dimensions of organizational readiness for change: commitment and efficacy. Technological changes in libraries, such as introducing VR/AR, require both high commitment and efficacy.

Librarians are part of the organization; thus, their readiness reflects their confidence in their ability to overcome the challenges imposed by change and their commitment to those changes. According to Rafferty *et al.* (2013), the measurement of cognitive and affective aspects should be differentiated in determining the readiness of librarians as members of an organization undergoing change.

The cognitive aspect describes the unique conditions for starting a change initiative, while the affective aspect describes a person's attitudes, beliefs and intentions. Cognitive factors associated with librarians' readiness to embrace technological change include alignment of ICT knowledge, skills and abilities. The affective aspect consists of conformity, how the librarian's beliefs change according to the demands imposed by their changing environment; library support, the individual's understanding of the commitment of leadership to the success of the transition; self-efficacy, the belief that individuals can adjust themselves to support organizational change; and the librarian's assumption that change will provide personal benefits (Rafferty *et al.*, 2013).

New technology adoption

Several researchers in the library literature have assessed technology readiness and explored the idea of an information technology adoption model by applying the Technology Readiness Index (TRI). According to Parasuraman and Colby (2015), the TRI has four variables: *Optimism* (OPT), a positive view of technology and the belief that it offers individuals and organizations increased control, flexibility and efficiency; *Innovativeness* (INV), the tendency to be a technology pioneer and opinion leader; *Discomfort* (DSC), a perceived lack of control over technology and feeling overwhelmed by it; and *Insecurity*

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(ISC), distrust of technology stemming from skepticism about its efficacy and concern about its potentially harmful consequences.

Optimism and innovation are contributors to technology readiness, while discomfort and insecurity are perceived as inhibitors of technology readiness. Discomfort decreases a user's proficiency and describes the need for more control over new technology, and insecure people do not trust the functionality and potential of technology (Parasuraman and Colby, 2015).

Research hypothesis

Library readiness for virtual reality/augmented reality adoption

In Newman and Newman's (2021) view, library management plays a crucial role in recognizing the impact of organizational change. Various studies have investigated library readiness to adopt new technology. Change readiness includes an assessment of library awareness of current changes, librarians' agility and well-defined change mechanisms. The results can show the level of organizational readiness to face change, but they cannot show whether the organization is ready to adopt new technology without resistance.

The TRI is a theoretical framework for measuring the readiness level for adopting new technology in library services (Parasuraman and Colby, 2015). Measuring library readiness helps determine its impact on the decision to adopt VR/AR by uncovering the interplay between organizational readiness and the TRI's inhibitors/contributors indicators. This reflects the assertion by Hradecky *et al.* (2022) that organizational readiness influences the contributor and inhibitor dimensions of adopting new technology, leading us to propose the following hypothesis:

H1. The readiness of libraries influences VR/AR adoption.

Librarian readiness for virtual reality/augmented reality adoption

Librarian readiness and librarians' roles and skills constantly evolve to suit social needs and technological developments (Shonhe, 2019). Individual readiness for change (IRFC) measures the cognitive and affective aspects.

Mahendrati and Mangundjaya (2020) tested the relationship between the IRFC and affective commitment to change through the TRI as a mediator variable. The results measured a strong relationship between IRFC and the TRI because the TRI provided a variable scale to determine the perception of readiness related to changes adopted at the individual level. Therefore, we propose the following hypothesis:

H2. The readiness of librarians influences VR/AR adoption.

Simultaneous library and librarian readiness for virtual reality/augmented reality adoption

The concept of organizational and individual readiness to change is part of change management theory. Change management includes planning, initiating, realizing, controlling and ultimately stabilizing the change process at the corporate and personal levels (Weiner, 2009). Libraries must change to survive, define their strategy and choose roles (Singha and Satpathy, 2017). The library's role is gradually changing from a traditional information warehouse to a technologically enabled service access provider. Therefore, we propose the following hypothesis:

H3. The readiness of libraries and librarians influences VR/AR adoption.

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Figure 1 provides a conceptual framework for our research hypotheses.

Methodology

This study used a quantitative approach to examine the influence of predetermined variables. The research population consisted of 740 librarians from university libraries, all members of the Indonesian Higher Education Library Forum (FPPTI). Slovin's formula was used to calculate the sample size, which, with a 5% margin of error and 95% confidence level, yielded 259. Random sampling was achieved by distributing a Google Forms link via email, WhatsApp groups and social media channels for each university library registered in FPPTI. The questionnaire was distributed twice within a three-month period; after the first distribution (December 2022 to January 2023) failed to reach the predetermined sample size, a second distribution (January to February 2023) was required.

Data collection

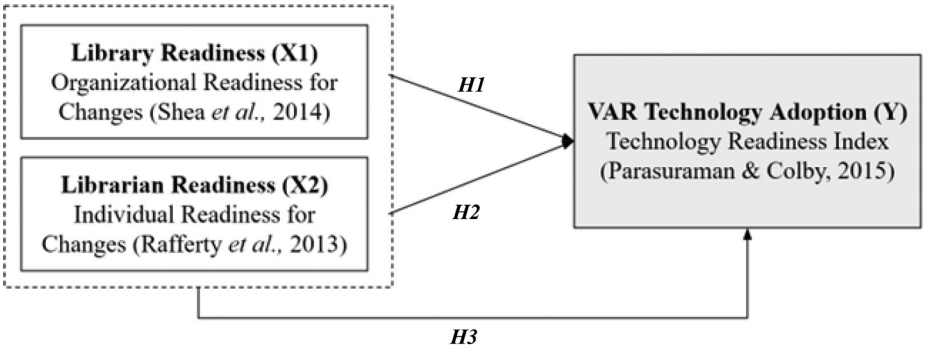
Primary data collection was via an online questionnaire using a Likert scale with 48 items measured based on ORIC theory (Shea *et al.*, 2014). The library readiness variable (X1) covered five questions on change commitment and seven questions on efficacy dimensions.

The librarian readiness variable (*X*2) included cognitive readiness, which was measured via 14 items based on IRFC theory consisting of knowledge, skills and abilities alignment (Rafferty *et al.*, 2013) and by adopting digital literacy theory to develop research question instruments consisting of ICT knowledge (Law *et al.*, 2018); and affective readiness, where five items were measured based on IRFC theory, comprising appropriateness, management support, self-efficacy and personal benefit (Rafferty *et al.*, 2013).

The readiness to adopt VR/AR in libraries variable (Y) included 17 items measured in four dimensions (optimism, innovativeness, discomfort and insecurity) based on the TRI theory (Parasuraman and Colby, 2015).

Data analysis

Frequency distribution was used to show data descriptions in the form of minimum, maximum, average, median, mode and standard deviation values for each variable of library readiness (X1), librarian readiness (*X*2) and VR/AR adoption in Indonesian university



Source: Author Modification from Shea *et al*. (2014), Rafferty *et al*. (2013), Parasuraman and Colby (2015)

Figure 1.
Conceptual framework

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libraries (Y). Category intervals were determined based on the total value of the variables obtained through the Likert scale criteria 1–5.

Testing for *H1* and *H2* examined the partial effect of library and librarian readiness on VR/AR adoption. In addition, *H3* was tested to evaluate the simultaneous effect of library and librarian readiness on VR/AR adoption. The coefficient of determination test was used to measure the magnitude of the simultaneous effect.

Results and discussion

The 259 respondents came from 34 public and 148 private university libraries in 17 Indonesian provinces, representing 70.27% of the predetermined sample number of FPPTI members. Most respondents were from West Java (23.2%), East Java (18.5%), Bali (15.4%), DKI Jakarta (11.6%) and Central Java (10%) (see Table 1).

Nearly two-thirds of respondents (59.5%) were women. The most popular age group was under 30 (35.1%), while another 24.7% were between 30 and 35. Most respondents (79.9%) were staff librarians, 36.3% had less than five years' experience and 29.3% had worked for 5–10 years. Most (68.3%) had earned undergraduate degrees.

To reveal the extent of VR/AR adoption, we designed the survey for all types of university libraries, regardless of whether they had already adopted VR/AR. As Figure 2 indicates, of the 182 university libraries that responded, 17 (9.4%) had already adopted some elements of VR/AR, 12 from private universities and five from public universities. Most of the remaining libraries (114, 62.6%) had yet to adopt VR/AR but were already planning to develop this capability. Just over a quarter (51, 28%) of university libraries still needed to formulate plans to adopt VR/AR.

Figure 3 illustrates the distribution of 17 university libraries already adopting VR/AR across three big islands in Indonesia: Java, Sulawesi and Sumatra. The highest reported percentage was from Java: five private university libraries in West Java and four in DKI Jakarta; one public and one private university library in Central Java; and one public and one private library in East Java. There was also one private university library in Banten, one public university in South Sulawesi, one private university in Central Sulawesi and one public library in North Sumatra.

The most popular VR/AR program (Figure 4) was a library tour, available in eight private universities (47% of the 17 libraries that had already adopted VR/AR programs). This accords with Sample (2020), who observed that library tours help students to become familiar with the library and can reduce students' or visitors' anxiety, especially on their first library visit. Virtual collections were available in three libraries (17.6%). Digital content exploration was reported in one private and one public university library (11.7%). Information literacy was available in three libraries (17.6%), while one private university offered an online catalog.

Findings

Libraries' readiness for virtual reality/augmented reality adoption

Table 2 demonstrates the libraries' readiness for change variable in Indonesian universities. A significantly high percentage of respondents (92.3%) agreed that their libraries were committed to adopting VR/AR following the needs and values of the library, closely followed by 90.7% of respondents who agreed that their libraries were committed to collaborating with various stakeholders, while 88.4% of respondents agreed that their library was committed to procuring a budget for adopting VR/AR, and 86.9% of respondents agreed that their library knows how it needs to prepare for adopting VR/AR. While still high, the lowest values were found for libraries committed to procuring a budget (84.9%). Based on respondents' views, university libraries in Indonesia have a strong commitment to adopting VR/AR aligned with needs,

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Characteristics	Frequency	%
Institution location		
Aceh	1	0.4
Bali	40	15.4
Banten	13	5.0
Bengkulu	1	0.4
DI Yogyakarta	11	4.2
DKI Jakarta	30	11.6
West Java	60	23.2
Central Java	26	10.0
East Java	48	18.5
West Kalimantan	2	0.8
Riau Island	1	0.4
Lampung	5	1.9
Maluku	1	0.4
East Sulawesi	5	1.9
Central Sulawesi	3	1.2
East Sumatra	2	0.8
North Sumatra	10	3.9
Gender		
Male	105	40.5
Female	154	59.5
Age		
<30	91	35.1
30–35	64	24.7
35–45	51	19.7
>45	53	20.5
Position		
Head Librarian	42	16.2
Head of Division	10	3.9
Librarian Staff	207	79.9
Years of experience		
<5	94	36.3
5–10	76	29.3
10–15	39	15.1
>15	50	19.3
Education		
Associate	35	13.5
Bachelor	177	68.3
Masters	43	16.6
PhD	4	1.5

Source: Authors' fieldwork (2023)

Table 1. Respondent demographic

collaboration and preparation, but this must be supported by a commitment to sufficient budget allocation to ensure sustained progress that their libraries know which parties must be involved in the adoption process. An equal number of respondents (79.1%) agreed that their library is capable of supporting librarians' adaptation processes and providing knowledge and skills in VR/AR. Furthermore, 78.5% of respondents agreed that their library can provide a clear workflow, and 74.5% of respondents agreed that their library is capable of establishing

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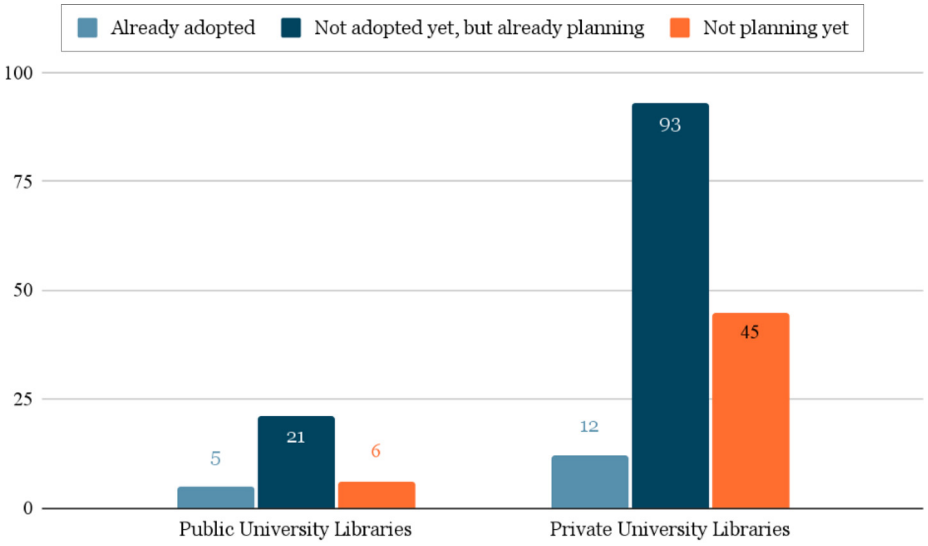


Figure 2.
VR/AR adoptions by university library type

Source: Figure by authors (2023)

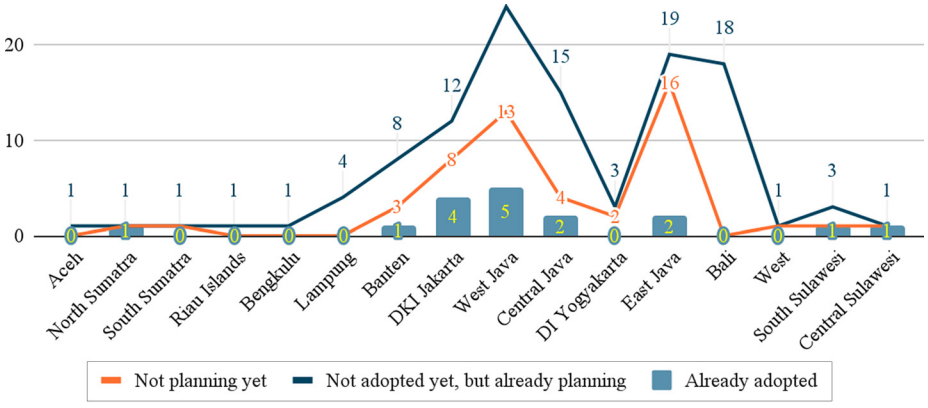


Figure 3.
VR/AR adoptions by location

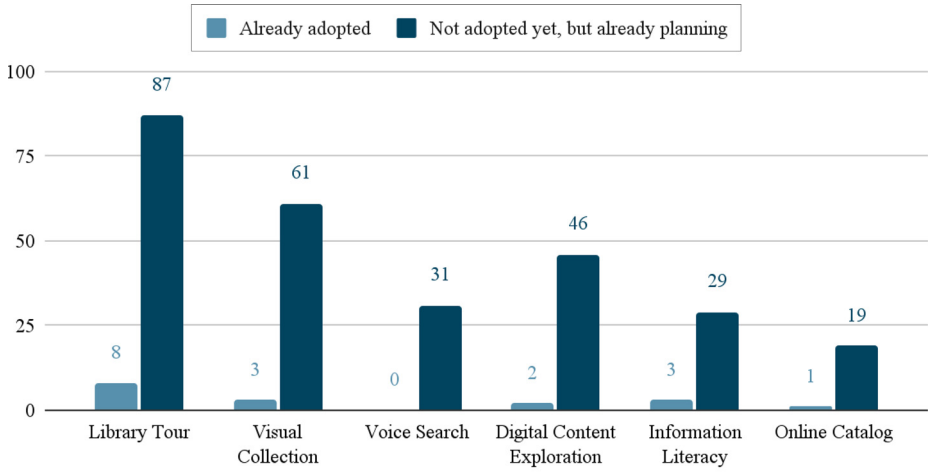
Source: Figure by authors (2023)

appropriate policies and regulations. The lowest values were found for the availability of an existing expert librarian to adopt VR/AR (64.4%) and the availability of an existing budget to adopt VR/AR (47.9%). This shows that while reinforcing support, training, clear workflows and policies, Indonesian university libraries should also support VR/AR adoption by prioritizing budget allocation and expertise.

Librarian readiness for virtual reality/augmented reality adoption

Table 3 presents the librarians' readiness variables in Indonesian university libraries. Cognitive aspects include knowledge, skills and ability alignment in digital literacy.

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Source: Figure by authors (2023)

Figure 4.
Type of VR/AR in university libraries in Indonesia

Reflecting the highest value in librarian knowledge, 68.8% of respondents were able to search relevant data and information with VR/AR; 67.2% of respondents had a global insight into digital technology development; and 60.6% had insight into national plans and targets for VR/AR adoption in libraries. The lowest values were found for their familiarity with VR/AR technology (54.1%); software and hardware (55.2%); and terms related to VR/AR (56%). Librarians in Indonesian universities show varying readiness for VR/AR, with strengths in data search but gaps in familiarity with technology types, software and terms.

Table 3 also shows the highest value in librarian skills, with 74.5% of respondents able to quickly understand the use of new technology. This is followed by 68.3% of respondents who can operate VR/AR applications and take advantage of commonly used features. The lowest values were found for their ability to map the relevance of VR/AR adoption (54.1%). This shows that librarians demonstrate strong skills in adapting to new technology and operating VR/AR applications proficiently, but face challenges in recognizing potential uses for VR/AR.

Lastly, the highest value in terms of librarian ability alignment (70.7%) was in respondents who indicated the ability to identify opportunities and challenges of VR/AR adoption to assist institutional decision-making. Conversely, the lowest value was connected to their understanding of the ethical drawbacks and consequences of presenting digital content in VR/AR (34%). This shows that university libraries in Indonesia should prioritize ethical training for librarians in VR/AR content presentation and enhance their understanding to effectively align with institutional decision-making, policies and regulatory compliance.

Regarding affective aspects (Table 4), 83% of respondents believed that technological development could improve performance at work. This is followed by 80.7% of respondents who believed that changes in VR/AR would provide personal benefits and 80.3% of respondents who believed that the adoption of VR/AR is in line with current technological developments. The lowest value (74.2%) is for respondents who believed that organizational leaders can successfully support technological development. Respondents expressed positive sentiments toward technological benefits at work, personal gains and alignment

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Code	Questions	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
Organizations' commitment (%)						
1	The library is committed to adopting VR/AR following the needs and values of the library	32.8	59.5	7.3	0.4	0
2	The library knows what to prepare for adopting VR/AR	34.4	52.5	11.6	1.5	0
3	The library is committed to procuring a budget for adopting VR/AR	36.3	48.6	13.5	1.5	0
4	The library is committed to collaborating with various stakeholders to adopt VR/AR	39.0	51.7	7.7	1.5	0
5	The library is committed to procuring qualified human resources for adopting VR/AR	41.3	47.1	10.4	1.2	0
Change efficacy (%)						
6	The library is capable of establishing appropriate policies and regulations	20.1	54.4	23.6	1.9	0
7	The library is capable of supporting librarians' adaptation process	24.7	54.4	20.5	0.4	0
8	The library is capable of providing a clear workflow	25.2	53.3	19.7	1.5	0
9	The library already knows which parties must be involved in the adoption process	23.9	57.1	17.4	1.5	0
10	The library already has an existing expert librarian to adopt VR/AR	21.2	42.9	27.0	7.7	1.2
11	The library is capable of providing knowledge and skills on VR/AR	26.6	52.5	18.1	1.9	0.8
12	The library already has an existing budget to adopt VR/AR	18.9	29.0	35.5	13.1	3.5

Notes: SA+A = agreed; SD+D = disagreed
Source: Authors' fieldwork (2023)

Table 2.
Libraries' readiness for VR/AR adoption

with technological trends, though leadership confidence lags. There is a need to encourage organizational leaders to boost confidence in technological success, ensure alignment with personal benefits and foster a positive perception of VR/AR adoption.

Virtual reality/augmented reality adoption readiness

The results in Table 5 demonstrate the variable readiness to adopt VR/AR. The dimensions of optimism and innovativeness, related to positive contributors, showed the highest value in that 99.2% of respondents were optimistic that adopting VR/AR would support library information literacy activities. In addition, in the innovativeness dimension, 64.1% of respondents agreed that VR/AR adoption in libraries can be developed with various innovations. These results align with Cook *et al.* (2019), who observed that university libraries are increasingly adopting VR technology for research and teaching purposes, including providing enhanced access to digital collections, offering new research tools and constructing new immersive learning environments for students. VR/AR offers university libraries in Indonesia enormous opportunities to provide new experiences to support information literacy, especially in digital scholarship.

However, libraries must also anticipate the challenges posed by the inhibitory dimensions of discomfort and insecurities when determining readiness to adopt VR/AR. Just under two-thirds of respondents (64.5%) felt there was inadequate data governance, while the highest value (76.9%) was for respondents reporting insecurity regarding the lack of qualified human resources for VR/AR adoption.

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Code	Questions	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
	Cognitive aspects (%)					
I	*Librarian knowledge*					
13	Familiar with VR/AR technology	22.4	31.7	6.9	22.4	16.6
14	Familiar with hardware and software commonly integrated with VR/AR	21.6	33.6	6.2	22.9	14.7
15	Familiar with terms related to the use of VR/AR	23.2	32.8	10.0	23.6	10.4
16	Able to search for relevant data and information with VR/AR	29.0	39.8	9.7	17.0	4.6
17	Have global insight into VR/AR development in the university library	26.3	40.9	9.3	19.3	4.2
18	Have insight into national plans and targets for VR/AR adoption in the library	25.9	34.7	10.4	22.4	6.6
II	*Librarian skills*					
19	Able to operate applications integrated with VR/AR and take advantage of commonly used features	33.6	34.7	8.9	15.1	7.7
20	Able to map the relevance of VR/AR adoption with library tasks and functions	28.6	35.1	10.0	17.0	9.3
21	Quickly understand the use of new technology	32.4	42.1	8.5	11.6	5.4
III	Librarian ability alignment					
22	Able to identify opportunities and challenges of VR/AR adoption to assist institutional decision-making	30.9	39.8	6.9	12.7	9.7
23	Able to control the risks of using VR/AR	20.8	38.6	10.4	19.3	10.8
24	Understand the ethics for presenting digital content in VR/AR	12.4	21.6	9.3	39.4	17.4

Notes: SA + A = agreed; SD + D = disagreed
Source: Authors' fieldwork (2023)

Table 3.
Cognitive aspects in librarian readiness

Influences of library and librarian readiness on virtual reality/augmented reality

Multiple linear regression analysis using SPSS determined the influence of the independent variables, library readiness (X1) and librarian readiness (*X*2) on VR/AR technology adoption readiness in university libraries (Y) simultaneously (*F*-test) and partially (*t*-test). The *F*-test results, presented in Table 6, show the significant value of the X1 and *X*2 effects on Y simultaneously was $0.000 < 0.05$, supporting *H3*, which means the readiness of libraries (X1) and librarians (*X*2) simultaneously influence the adoption of VR/AR technology in university libraries in Indonesia (Y).

A partial *t*-test was used to examine the influence of the independent variables partially on the dependent variable. The results, shown in Table 7, reveal that the significance value for the library readiness variable (X1) was $0.139 > 0.05$. Because the calculated significance value is above 0.05, this result does not support our hypothesis that library readiness affects VR/AR adoption in university libraries in Indonesia. Therefore, *H1* is rejected.

On the other hand, the librarian affective readiness variable (*X*2) had a significant value of 0.000, which was less than 0.05, leading the readiness of librarians to influence VR/AR adoption in university libraries in Indonesia (*H2*) to be accepted. Adopted or planned VR/AR by university libraries requires the readiness of librarians as professional individuals to understand, accept, use and quickly adapt to this new

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Code	Questions	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
		Affective aspects (%)				
27	Believes that the adoption of VR/AR is in line with current technological development	37.1	43.2	8.1	8.1	3.5
28	Believes that organizational leaders are capable of succeeding and supporting technological development	29.0	45.2	5.8	10.8	9.3
29	Believes in the ability to accept technological development in work	30.9	48.6	6.6	9.7	4.2
30	Believes that technological development would provide personal benefits	34.4	46.3	6.9	8.9	3.5
31	Believes that technological development could improve performance at work	37.8	45.2	6.9	8.5	1.5

Notes: SA + A = agreed; SD + D = disagreed
Source: Authors' fieldwork (2023)

Table 4.
Affective aspects in librarian readiness

technology. These observations are in line with the assertion by Rafferty *et al.* (2013) that measuring librarian readiness to change should include both cognitive (knowledge, skills and ability alignment) and affective (suitability for changes, management support, self-efficacy and personal benefit) aspects.

The coefficient of determination coefficient, which has a value in the range of 0 to 1, can be used to quantify the relationship between several variables. A value closer to 0 implies that the independent variables are very limited in explaining the variation of the dependent variable. However, a value closer to 1 or 100% implies that the independent variables provide most of the information needed to predict the variation in the dependent variable.

As can be seen in Table 8, the adjusted R^2 value was 0.909, supporting the conclusion that the simultaneous effect of library and librarian readiness on the adoption of VR/AR in university libraries was 90.9%. Only 0.091 or 9.1% was influenced by factors outside the library and librarian readiness variables.

Implications

There are practical implications for library institutions in Indonesia in terms of understanding the attendant opportunities and challenges when preparing organizational ecosystems. These findings provide a broad overview of VR/AR adoption in university libraries in Indonesia and contribute to understanding the specific needs and interests of universities in Indonesia considering adopting VR/AR. In addition to preparing adequate infrastructure, libraries should ensure the availability of adequate human resources. Based on the challenges identified in adopting VR/AR in university libraries in Indonesia, several recommendations can be made:

(1) Library readiness:

- Prioritize and commit to allocating dedicated budgets for the adoption of VR/AR. This financial commitment is crucial to overcoming budget-related challenges.

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Table 5. VR/AR adoption readiness

Code	Questions	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
Optimism (%)						
32	Adopting VR/AR in the library will benefit the entire community	32.8	61.4	4.6	1.2	0
33	Adopting VR/AR can support the efficiency and effectiveness of library services	30.5	65.3	2.7	1.5	0
34	Adopting VR/AR can support virtual library tour	33.2	64.1	2.3	0.4	0
35	Adopting VR/AR can support visual information that directs the required collections	35.9	59.1	3.9	0.4	0.8
36	Adopting VR/AR can enable a voice search	29.0	62.2	4.2	2.3	2.3
37	Adopting VR/AR can provide virtual digital content exploration in a real environment	29.7	66.4	2.7	0.4	0.8
38	Adopting VR/AR can support library information literacy activities	35.1	64.1	0.4	0.4	0
39	VR/AR can be integrated with the library's online catalog system	32.8	63.7	3.1	0.4	0
Innovativeness (%)						
40	Tendency to be a pioneer and thought leader in VR/AR adoption	11.2	52.5	17.8	10.4	8.1
41	VR/AR adoption in libraries can be developed with various innovations	12.0	52.1	15.8	11.2	8.9
Discomfort (%)						
42	Lack of collaboration between actors to attract talent and support	7.3	52.9	15.1	15.8	8.9
43	Inadequate data governance	11.6	52.9	15.1	15.4	5.8
Insecurity (%)						
44	Lack of human resources with qualified skills	14.7	62.2	11.2	8.9	3.1
45	Risks in adopting VR/AR in libraries	6.2	49.4	18.1	15.4	10.8
46	Digital divide	9.7	51.4	15.1	13.5	10.4
47	Technical challenges	13.9	56.0	14.7	12.0	3.5
48	Regulatory alignment related to adopting VR/AR in libraries	13.1	51.7	17.8	11.6	5.8

Notes: SA+A = agreed; SD+D = disagreed
Source: Authors' fieldwork (2023)

- Develop human resources: Address the shortage of qualified personnel by investing in training programs. This will enhance the skills of library staff, particularly in ICT, to effectively navigate VR/AR adoption.

(2) Librarian readiness:

- Design and implement comprehensive training programs for librarians to improve both cognitive and affective aspects related to VR/AR adoption. This will ensure that librarians have the necessary skills and knowledge.
- Offer organizational support to librarians to facilitate their professional development, especially in the context of emerging technologies such as VR/AR. This support can include mentorship, workshops and ongoing learning opportunities.

(3) Readiness to adopt VR/AR:

- Foster collaboration with various stakeholders, including librarians, administrators and external partners, to attract talent and garner support for VR/AR adoption. Collaborative efforts can address challenges more effectively.
- Improve data governance: Address concerns related to data governance by establishing clear policies and frameworks for managing VR/AR-related data. Ensuring data security and privacy is crucial for successful adoption.

Table 6.
Simultaneous effect (*F*-test)

ANOVA[a]						
Model		Sum of squares	df	Mean squares	*F*	Sig.
1	Regression	6,135.085	2	3,067.542	74.544	0.000[b]
	Residual	10,534.583	256	41.151		
	Total	16,669.668	258			

Note: [a]Dependent variable: VR/AR adoption readiness; [b]Predictors: (constant), library readiness
Source: Authors' fieldwork (2023)

Table 7.
Partial effect (*t*-test)

Coefficients[a]						
Model		*B*	Std. error	Beta	*t*	Sig.
1	(Constant)	31.594	3.093		10.215	0.000
	Library readiness	0.103	0.070	0.095	1.484	0.139
	Librarian readiness	0.418	0.049	0.543	8.501	0.000

Note: [a]Dependent variable: VR/AR adoption readiness
Source: Authors' fieldwork (2023)

Table 8.
Coefficient of determination

Model summary				
Model	*R*	*R* squared	Adjusted *R* squared	Std. error of the estimate
1	0.954[a]	0.910	0.909	1.43583

Note: [a]Predictors: (constant), library readiness
Source: Authors' fieldwork (2023)

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- Address human resource insecurity: Develop strategies to mitigate concerns about the availability of qualified human resources. This may involve creating partnerships with educational institutions or providing incentives for professionals to join the library workforce.

By implementing these suggestions, university libraries in Indonesia can enhance their readiness for VR/AR adoption, overcome challenges and create a conducive environment for the effective integration of these technologies into academic activities.

The findings also provide new insight into the use of ORIC, IRFC and TRI in libraries' adoption of new technology. First, the significant influence of librarians' readiness to adopt VR/AR in university libraries in Indonesia aligns with the assertions of Shonhe (2019) and Mahendrati and Mangundjaya (2020) that individual readiness to change influences new technology adoption. Also in line with Mahendrati and Mangundjaya (2020), this research highlights the significant influence of IRFC (Rafferty *et al.*, 2013) on TRI (Parasuraman and Colby, 2015), which can be used to measure librarian readiness to adopt new technology.

Furthermore, even though this research has not proved the influence of library readiness on VR/AR adoption, the simultaneous readiness of libraries and librarians was found to influence VR/AR adoption in Indonesian university libraries. These results support Hradecky *et al.*'s (2022) finding that organizational readiness to change, including size, financial, infrastructure and strategic plans, influences readiness to adopt new technology, which, in turn, influences adoption *by people* of new technology; thus, it cannot be separated from the availability of human resources. Therefore, it can be concluded that librarian readiness is a part of library readiness and is a mutually supportive unit.

Conclusion

The adoption of VR/AR among university libraries in Indonesia has not yet reached the point of significant growth. According to our findings, 12 private and five public university libraries, mostly located in Central Java, have adopted this technology. Java is an island in Indonesia where the capital city of Jakarta is located. In fact, most of the reputable universities in Indonesia are located on the island of Java. With a variety of academic opportunities, Java attracts students both nationally and internationally, creating a rich educational environment with various development efforts, including the adoption of new technologies in libraries. The most widely adopted VR/AR application by Indonesian university libraries is the 360-degree virtual library tour, but there is great interest in adopting more VR/AR applications. This study examined universities across 16 Indonesian provinces, and all exhibited interest in adopting VR/AR in the future.

Challenges to adopting VR/AR are the extent of libraries' commitment to procuring a budget and the lack of qualified human resources. Librarians' cognitive and affective aspects require improvement through training programs and the availability of organizational support to provide skills development, especially in the ICT field, to deal with the adoption of VR/AR in libraries.

Respondents were optimistic and willing to innovate to increase the effectiveness and efficiency of academic activities, such as digital scholarship. However, they also indicated discomfort regarding the lack of collaboration between actors to attract talent and support VR/AR adoption, inadequate data governance (in the high category) and insecurity regarding the lack of qualified human resources.

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The research also revealed the simultaneous influence of the readiness of libraries and librarians on VR/AR adoption readiness in university libraries in Indonesia. In part, librarians' readiness influenced VR/AR adoption significantly and positively. Contrariwise, library readiness did not influence VR/AR adoption readiness. Thus, librarians' readiness cannot be separated from library readiness. The readiness of librarians as personnel directly affected by relentless technological developments and library readiness for adopting VR/AR are integral parts of library management that must support each other.

Limitations and recommendations

This study only tested the readiness of VR/AR adoption in university libraries in Indonesia. Thus, future researchers might consider assessing the readiness of librarians and libraries for adopting VR/AR within school, special and national libraries, or in other countries. In addition, further research can assess how user needs might influence librarians' decisions around adopting VR/AR.

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